

Machine Learning and Data Analytics Sample Questions

UNIT – I : Machine Learning Fundamentals & Case Studies

- 1. Differentiate between supervised and unsupervised learning** with suitable real-life examples.
Explain how classification and regression differ in supervised learning.
- 2. Explain the concept of dimensionality reduction.**
Why is it important in machine learning?
Illustrate your answer with an example from real-world datasets.
- 3. What is clustering?** Explain how clustering differs from classification.
Discuss the importance of clustering in business analytics.
- 4. Explain the working of K-Means clustering algorithm.**
What are its advantages and limitations?
- 5. Discuss the Balanced Scorecard Model** for measuring organizational performance.
How can machine learning techniques enhance performance evaluation?
- 6. Explain the Employee Attrition Prediction problem.**
How can classification algorithms help organizations reduce attrition rates?
- 7. Describe the Market Capitalization Categories case study.**
How can clustering or classification be used to categorize companies based on market value?
- 8. Explain the Performance Appraisal System** in organizations.
How can data analytics improve decision-making in employee performance evaluation?
- 9. Explain the Technology Acceptance Model (TAM)** and its role in evaluating cloud computing adoption.
- 10. Describe the Customer Buying Intention Prediction problem.**
How does digital marketing data help in predictive modeling?

UNIT – II : Supervised Learning Algorithms

- 11. Explain the Naïve Bayes classifier.**
Why is it called “naïve”? Discuss its advantages and limitations.
- 12. Using an example, explain how Naïve Bayes can be used for email spam classification.**
- 13. Describe the K-Nearest Neighbour (KNN) algorithm.**
How does the value of K affect classification performance?
- 14. Given a dataset with attributes (Maths, Science, Result), explain step-by-step how KNN classifies a new data point using Euclidean distance.**

15. Explain the **Support Vector Machine (SVM)** algorithm.
What is the significance of margin and support vectors?
16. Differentiate between **linear SVM and non-linear SVM**.
Explain the role of kernel functions.
17. Explain the working of **Decision Trees**.
What is entropy and information gain?
18. What is overfitting in decision trees?
Explain pre-pruning and post-pruning techniques.
19. Discuss the case study “**Predicting Phishing Websites**” using machine learning techniques.
20. Explain the **Fraud Detection in Credit Card and Mobile Payments** case study.
Which ML algorithms are suitable and why?

UNIT – III : Data Analytics & Statistical Foundations

21. Explain the relationship between **Data Science, Data Analytics, and Big Data Analytics**.
22. Describe the major components of Data Science.
Explain the role of data scientists in organizations.
23. Define and explain:
- Mean
 - Median
 - Mode
 - Variance
 - Standard Deviation
- Give suitable examples.
24. What is a probability distribution?
Differentiate between discrete and continuous probability distributions.
25. Explain hypothesis testing.
Differentiate between null hypothesis and alternative hypothesis.
26. Describe sampling techniques and explain their importance in data analysis.
27. Explain the concept of statistical inference and its importance in decision-making.
28. Discuss the use of statistical measures in business analytics with real-world examples.

UNIT – IV : Data Preprocessing, Visualization & Case Studies

29. Explain data preprocessing.
Discuss methods for:
- Handling missing values

- Data normalization
- Data transformation

30. Differentiate between data preprocessing, data reduction, and feature extraction.

31. Explain univariate and multivariate data analysis with suitable examples.

32. Describe the concept of **Gradient Descent** and its role in machine learning optimization.

33. Explain the role of **data visualization** in data analytics.
Compare **Matplotlib** and **ggplot2**.

34. Discuss the **Insurance Policy Offer Prediction** case study.
How does data analytics help in targeting potential customers?

35. Explain the **Discount Targeting in Online Shopping** case study.
How can customer segmentation improve marketing strategies?

36. What are the steps involved in building a complete data analytics solution?

37. Discuss ethical issues in data analytics such as data privacy, bias, and fairness.

38. Explain how exploratory data analysis (EDA) helps in understanding datasets before model building.

39. Compare supervised and unsupervised learning with respect to:

- Data requirement
- Output type
- Applications

40. Describe how machine learning can be applied in real-world domains such as healthcare, finance, and e-commerce.

41. Explain the importance of model evaluation metrics such as accuracy, precision, recall, and F1-score.

42. Discuss challenges faced in real-world data analytics projects and possible solutions.

43. Explain the complete machine learning pipeline starting from data collection to deployment.

44. Describe how data visualization helps decision-makers understand complex datasets.

45. Explain the role of machine learning in predictive analytics with suitable examples.

46. Discuss how bias and variance affect model performance.

47. Explain the importance of feature engineering in building accurate models.

48.What are the ethical considerations in data analytics and AI-based systems?

49.

Explain how machine learning can support business intelligence and strategic planning.

50.Write short notes on:

- a) Supervised vs Unsupervised Learning
- b) Data Mining vs Data Analytics
- c) Model Evaluation Techniques
- d) Role of ML in Digital Transformation